AUTOMATED PARKING SYSTEM WITH LICENCE PLATE RECOGNITION AND SLOT ALLOCATION

Submitted in partial fulfillment of the requirements for the award of
Bachelor of Engineering degree in Computer Science and Engineering
with Artificial Intelligence and Machine Learning

By

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BONAFIDE CERTIFICATE

This is to certify that this Project Report is the bonafide work of **PRANAY JAVVAJI (42611097), PUSHPARAJ BUVIGNESH (42611103)**, who carried out the Project entitled "AUTOMATED PARKING SYSTEM WITH NUMBER PLATE RECOGNITION AND SLOT ALLOCATION" under my supervision from November 2025 to April 2026.

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I, **PRANAY JAVVAJI (Reg. No- 42611097),** hereby declare that the Project Report entitled **"AUTOMATED PARKING SYSTEM USING NUMBER PLATE RECOGNITION AND SLOT ALLOCATION"** done by me under the guidance of **Dr. Dr. MINU SUSAN JACOB, M.E., Ph.D.,** is submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering degree in **Computer Science and Engineering with Artificial Intelligence and Machine Learning.**

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ABSTRACT

Automated parking has become an essential requirement in modern urban environments, where space constraints and increasing vehicle numbers make manual parking inefficient and error-prone. Traditional parking systems require human supervision and often lead to mismanagement, time wastage, and an increased risk of collisions during tight parking maneuvers. Despite technological progress, integrating multiple intelligent functions such as vehicle detection, license plate recognition, collision prevention, and slot allocation into a single automated system remains a challenge. This study proposes an intelligent parking management system that leverages YOLO-based object detection for classifying vehicles as cars or bikes and EasyOCR for accurate license plate recognition. The system dynamically allocates available parking slots and maintains a real-time database storing vehicle details, slot numbers, and entry timestamps. At exit, it retrieves the record, calculates parking duration, and computes the corresponding fare. Furthermore, vehicle re-identification ensures continuous tracking throughout the parking process, while a camera calibration-based collision detection module estimates distances to walls and pillars, reducing accidents during parking. Through the integration of these technologies, the proposed system aims to enhance efficiency, safety, and accuracy in parking operations, minimize human intervention, and provide a scalable solution for smart cities, commercial complexes, and public parking facilities, ultimately supporting the vision of intelligent transportation systems.

TABLE OF CONTENTS

Chapter No	Title	Page No.
	ABSTRACT	**v**
	LIST OF ABBREVIATIONS	**viii**
	LIST OF FIGURES	**xi**
	LIST OF TABLES	**x**
1	**INTRODUCTION**	**1**
	1.1 Background and Motivation	1
	1.2 Importance Of This Project	2
	1.3 Objectives	**3**
	1.4 Overview Of The Proposed System	4
	1.5 Role of Deep learning and YOLOv8	4
2	**LITERATURE SURVEY**	**6**
	2.1 Inference from literature survey	12
	2.2 Open problems in Existing System	13
3	**AIM AND SCOPE OF THE PRESENT INVESTIGATION**	**15**
	3.1 Aim Of The Project	15
	3.2 Scope Of The Present Investigation	16
	3.3 Novelty Of The Proposed System	17
	3.4 System Requirements	18
4	**ALGORITHMS AND METHODOLOGY**	**21**
	4.1 System Architecture Overview	21
	4.2 Application Flow	21
	4.3 Intelligent Processing Models	26
	4.4 Algorithm Logic and Control Flow	28

4.5 Data Architecture and Mathematical Ingestion 29

4.6 User Interface and System Configuration 32

5 **RESULTS AND DISSCUSSION, PREFOMANCE ANALYSIS 34**

5.1 Performance of the Custom Number plate Detection Model 34

5.2 Evaluation of Pre-Trained Vision Models 40

5.3 Vehicle Re-Identification Performance 43

5.4 Spatial Occupancy and Tracking Accuracy 46

5.5 End-to-End System Latency 47

6 **SUMMARY AND CONCLUSIONS 49**

6.1 Summary of the Project 49

6.2 Conclusion and Final Reflections 50

6.3 Technical Limitations and Challenges 50

6.3 Future Work and Scalability 51

REFERENCES 54

Annexure 1: Plagiarism Report 55

Annexure 2: Conference Paper Submission 56

Annexure 3: SDG Mapping 58

Annexure 4: PO PSO Mapping 60

Annexure 5: Additional Screenshots 62

LIST OF ABBREVATIONS

ABBREVATION	EXPANSION
ANPR	Automatic Number Plate Recognition
API	Application Programming Interface
CNN	Convolutional Neural Network
CRR	Character Recognition Rate
CUDA	Compute Unified Device Architecture
DFL	Distribution Focal Loss
FAR	False Acceptance Rate
FPS	Frames Per Second
GPU	Graphics Processing Unit
IoU	Intersection over Union
JSON	JavaScript Object Notation
LPR	License Plate Recognition
OCR	Optical Character Recognition
Re-ID	Vehicle Re-Identification
RFID	Radio Frequency Identification
RTSP	Real-Time Streaming Protocol
UI	User Interface
V2I	Vehicle-to-Infrastructure
YOLO	You Only Look Once

LIST OF FIGURES

Figure No	Figure Name	Page No.
4.1	System Architecture	24
4.2	Entity-Relationship (ER) Diagram of the centralized SQLite database	31
5.1	YOLOv8 Training and Validation Metrics over 25 Epochs.	35
5.2	Pression-Recall (PR) Curve achieving 0.945 mAP@0.5.	36
5.3	F1-Confidance Curve showing peak optimal threshold.	37
5.4	Normalized Confusion Matrix for YOLOv8 Number Plate Detection	38
5.5	Real-Time Vehicle Detection (YOLOv8)	41
5.6	Number Plate Text Extraction	43
5.7	Cosine Similarity Graph	45

LIST OF TABLES

Table No.	Table Name	Page No.
3.1	Comparison of Proposed System with Existing Systems	17
3.2	Hardware System Specifications	19
3.3	Software Stack and Frameworks	20
5.1	Custom YOLOv8 Number Plate Detection Performance.	38
5.2	Cosine Similarity Matching Matrix	46
5.3	Inference Latency Breakdown per Vehicle Event	48